

Esercizio.

Let L fol, M B -valued model, B a well behaved b.a. and $\forall \phi \in L(M)$ $D_\phi := \{G \in St(B) : M/G \models \phi\}$. It is not in general true that D_ϕ dense in N_b implies $b \leq [[\phi]]$.

Dimostrazione.

Consider $L = \{E\}$, $M = \omega$, $B = RO(2^{\omega \times \omega})$ and put $[[m = n]] = 1$ if $m = n$ and 0 otherwise. Put $[[E(m, n)]] := U_{m,n}$ where $U_{m,n} = \{f \in 2^{\omega \times \omega} : f(m, n) = 1\}$. One can check that is is a well behaved B -valued model. Now consider $\phi : \exists x \forall y (x \neq y \rightarrow E(x, y))$.

- $[[\phi]] = 0$: in fact $[[\phi]] = \bigvee_{n < \omega} \bigwedge_{m \neq n} [[E(m, n)]]$. Now observe that for all $n < \omega$ we have $\bigwedge_{m \neq n} [[E(m, n)]] = Reg(\{f \in 2^{\omega \times \omega} : \forall m \neq n f(m, n) = 1\})$. Since $\{f \in 2^{\omega \times \omega} : \forall m \neq n f(m, n) = 1\}$ is closed and has empty interior we get $\bigwedge_{m \neq n} [[E(m, n)]] = 0$.
- D_ϕ dense in $N_{RO(2^{\omega \times \omega})} = St(B)$: in fact an open set in $St(2^{\omega \times \omega})$ must have as a subset an element of the form N_A for some $A \in RO(2^{\omega \times \omega})$. Now, since in particular A is open subset of $2^{\omega \times \omega}$, there must be $U_{n,m} \subseteq A$. Then it's enough showing $D_\phi \cap N_{U_{n,m}} \neq \emptyset$. So we must find $G \in St(B)$ such that $U_{n,m} \in G$ and such that $M/G \models \phi$. Now $M/G \models \phi$ means that there is some $r < \omega$ such that for all $s \neq r$ we have $[[E(s, r)]] \in G$ (that is $U_{s,r} \in G$). Consider $\{U_{n,s} : n \neq s\}$. This is a prefilter, so extends to an ultrafilter. That is the ultrafilter we were looking for.

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